

QIANHOU QU

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RESEARCH INTERESTS

My research interests focus on device-circuit-architecture co-design for emerging-memory-based and neuromorphic computing systems, with an emphasis on variation-aware modeling, SPICE-level validation, and machine-learning-assisted compact modeling for analog/mixed-signal simulation acceleration.

EDUCATION

University of Texas at Arlington

Arlington, TX

Ph.D. Student in Electrical Engineering, GPA: 4.00/4.00

Jan. 2025 - Present

University of Southern California

Los Angeles, CA

M.S. in Electrical Engineering (VLSI Design), GPA: 3.82/4.00

Aug. 2023 - Dec. 2024

- Relevant coursework: MOS VLSI Design (ranked 8/127), Computer System Organization, VLSI System Design, Computer System Architecture

Hebei University of Technology

Tianjin, China

B.E. in Automation

Aug. 2018 - Jul. 2022

RELATED PROJECTS

Memristor-Based In-Memory Computing Accelerator for Spiking Neural Networks

Aug. 2025 - Apr. 2026

- Designed a memristor-based neuromorphic accelerator for a bio-inspired predator-prey interception task using a spiking neural network architecture.
- Developed a 1T1R memristive crossbar and integrate-and-fire neuron interface for in-memory synaptic accumulation and asynchronous spike-based inference.
- Implemented SPICE-level circuit models of memristive synapses, neuron integrators, comparators, reset logic, and spike-generation interfaces using the SkyWater SKY130 PDK.
- Established a read-noise-aware hardware mapping flow to deploy software-trained SNN parameters onto multi-level memristive conductance states.
- Validated circuit-to-system inference accuracy against the software SNN baseline, achieving a correlation coefficient of 0.9622, an MAE of 0.0544, and a 96% interception success rate.
- Achieved 10.67 pJ/spike energy consumption and 906 μm^2 neuron-level layout area.

Neural-Network-Assisted Compact Verilog-A Modeling for Analog Circuit Simulation

May 2026 - Present

- Developed a neural-network-assisted compact modeling framework to accelerate transistor-level analog circuit simulation using SPICE-generated data.
- Built HSPICE/SKY130 simulations for comparator, op-amp, and integrate-and-fire neuron circuits, generating waveform datasets across varying inputs, loads, and timesteps.
- Trained compact neural networks to predict circuit state evolution from operating conditions and exported the trained models as Verilog-A behavioral modules for SPICE-level simulation.
- Validated accuracy against transistor-level references through waveform, membrane-potential, and pulse-timing comparisons.
- Demonstrated the model in large-scale integrate-and-fire neuron simulations, achieving approximately $10\times$ runtime speedup over full transistor-level simulation.

Bio-Inspired Predator-Prey Training with STDP+RL

Jun. 2025 - Aug. 2025

- Developed a bio-inspired SNN for predator-prey interception using STDP combined with reinforcement learning (RL) for reward-driven synaptic adaptation.
- Implemented and trained the network in Brian2 by encoding relative trajectories into spike trains and evaluating closed-loop pursuit behavior.
- Mapped the trained network to memristive crossbar synapses and analog integrate-and-fire neuron circuits in SPICE,

with weight-to-conductance mapping and in-memory accumulation.

Variation-Aware Memristor-Based In-Memory Computing Accelerator for ViT

Jan. 2025 - Jun. 2025

- Developed a variation-aware modeling framework for memristor-based in-memory computing (IMC) applied to Vision Transformer inference.
- Built device-to-algorithm mapping models to analyze the impact of memristor conductance precision, device variation and analog computation noise.
- Integrated hardware non-idealities into TIMM-based ViT inference to evaluate ImageNet accuracy degradation and robustness under hardware constraints.

PUBLICATIONS

- **Q. Qu**, S. Lu, J. Shang, S. Jung, Q. Liang, and C. Pan, "Fast and Energy-Efficient Analog Accelerator for Vision Transformer," 2025 IEEE 68th International Midwest Symposium on Circuits and Systems (MWSCAS), 2025.
- **Q. Qu** and C. Pan, "Memristor-Based Spiking Neural Network Accelerator for Neuromorphic Computing," ECI Workshop on Neuromorphic Computing, 2026.
- **Q. Qu**, S. Lu, L. Shang, J. Utailawon, S. Jung, Q. Liang, and C. Pan, "Memristor-Based Spiking Neural Network Accelerator for Bio-inspired Interception Task," accepted by 2026 IEEE Dallas Circuits and Systems Conference (DCAS), 2026.
- **Q. Qu**, S. Lu, L. Shang, S. Jung, Q. Liang, and C. Pan, "Variation-Aware Memristor-Based Analog Accelerator for Vision Transformer," Electronics, vol. 15, no. 5, article 1116, 2026.
- **Q. Qu**, S. Lu, S. Jung, Q. Liang, and C. Pan, "Compact and Energy-Efficient Memristive Spiking Neuromorphic Accelerator for Bio-inspired Interception Tasks," under review at ACM Journal on Emerging Technologies in Computing Systems (ACM JETC), 2026.

TEACHING EXPERIENCE

Teaching Assistant, Electrical Engineering Laboratories, University of Texas at Arlington

- Delivered laboratory lectures and pre-lab instructions on circuit experiments, measurement procedures, and instrumentation usage.
- Guided undergraduate students through hands-on experiments, equipment setup, troubleshooting, and data collection.
- Supported office hours, lab report grading, feedback, and student communication.

SKILLS

- **Circuit Design:** Analog/Mixed-Signal IC Design, SRAM, Memristor Crossbar, IMC Architectures.
- **Simulation Tools:** Cadence Virtuoso, Spectre, HSPICE, ngspice.
- **Programming:** MATLAB, Python.
- **Modeling:** Device Variation Modeling, SPICE Simulation, PPA Analysis, compact modeling, integrate-and-fire neuron circuits.